Department of Computer Science and Engineering (CSE)
BRAC University

CSE251 – Electronic Devices and Circuits

# INTRODUCTION TO ELECTRONICS

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Prepared by Utsab Saha [UTS]

# Lecture 01: Outline

- Course Prologue
- History of Electronics
- Revisit to CSE 250
- Circuit Schematics & Representation
- Alternative Circuit Representation: Line Diagram

# Course Prologue

## Timelines:

| | |
|---|---|
| 31.01.2026 | Classes of Spring begin |
| 30.03.2026 | MID Term Exam |
| 10.05.2026 | Last class of Spring 26 |
| 13.05.2026 | Final Exam |

# Course Prologue

Distribution of Marks:

| Assessment | Percentage | Total number of assessments | Number of assessment to be graded |
|---|---|---|---|
| Attendance | 5% | - | - |
| Assignment | **5%** | 3/4 | [if n>3; Best (n-1)] |
| Quiz | 15% | 4 | Best 3 [Best (n-1)] |
| Midterm | 25% | 1 | 1 |
| Final | 25% | 1 | 1 |
| Lab | 25% | - | - |

# Course Prologue

## Things to remember:

- Quiz questions should help prepare the students for the midterm and final exams.
- Quiz, midterm, and final may contain bonus questions, but that will be at most 10% of the total marks of the assessment.
- Questions for quiz, midterm, and final are often modified versions of assignments
- You can collab, but you cannot copy. Plagiarism will result in null marks.

# Course Prologue

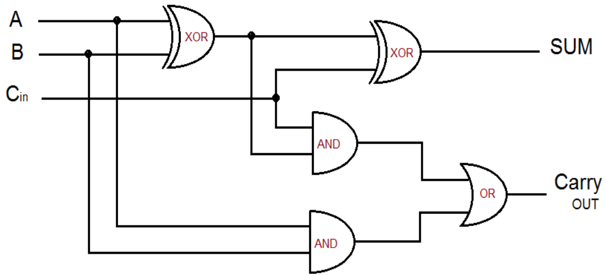
- Absence/Late Policies:
  - Attendance will be recorded and shared
  - Attendance: P/A/L. You can be ‘Excused’ for ‘A’ if you show Medical documents
  - **Attendance<70%** won’t qualify for Mid/Final
  - Assignment deadlines won’t change
  - You can be late for a total of 4 days for assignments in the entire semester
- **DO NOT**:
  - Copy/Cheat. If so, negative/capped marks/suspension will be the outcome

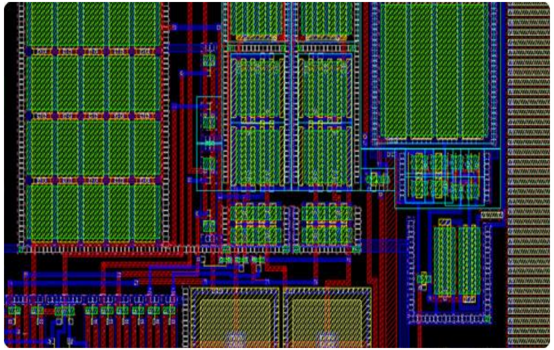
# History of Electronics

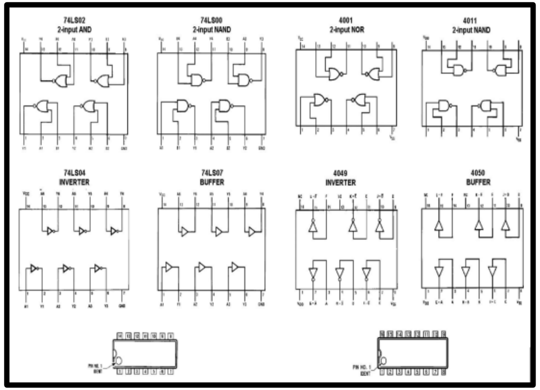
- Electronics emerged with the discovery of the electron in 1897 and the invention of the vacuum tube, which amplified and rectified electrical signals.
- Vacuum tubes [Link] were the first active electronic components and revolutionized various industries, including radio, television, telephony, and music recording.
- The point-contact [Link] transistor was invented in 1947, marking a significant technological advancement, although vacuum tubes still dominated until the 1980s.
- The IBM 608 [Link] calculator in 1955 became the first commercial product to use transistors exclusively, leading to their widespread use in computer logic and peripherals.
- The MOSFET [Link], invented in 1959, revolutionized the electronics industry with its compact size, mass production capabilities, low power consumption, and versatility.
- The integrated circuit/IC, developed by Jack Kilby and Robert Noyce, solved the problem of circuit size and speed by integrating components onto a single semiconductor block.
- This led to advancements in small-scale integration (SSI), medium-scale integration (MSI), and very large-scale integration (VLSI), with billion-transistor processors becoming available in 2008.

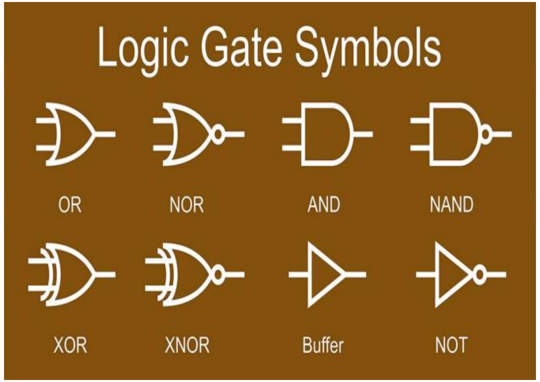
# Why do we need Electronics?

1. Switches/Logics
2. Arithmetic Operations
3. IC design
4. Chip Design
5. **Computers**

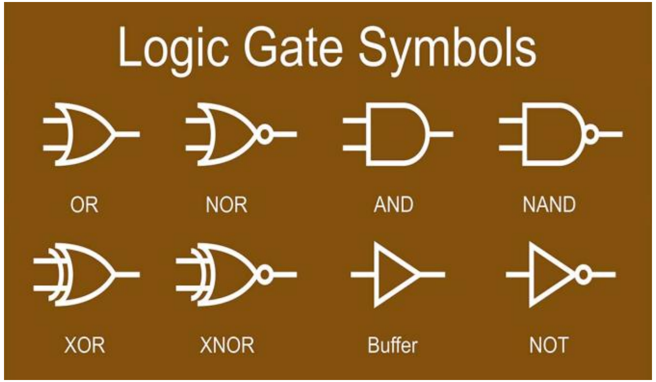


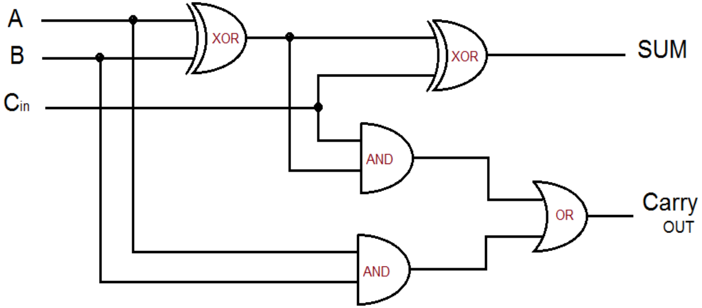



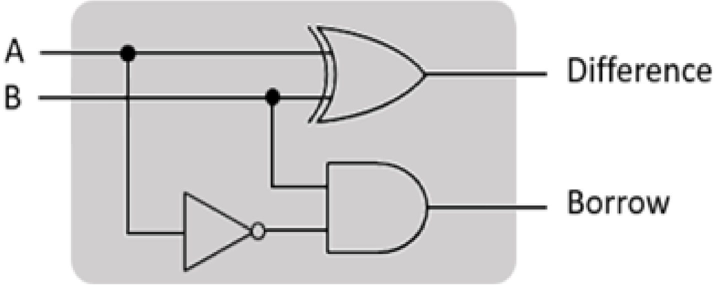


# Background




- The initial task of the Computer was to 'Compute'. To be more specific, it made mathematical operations automatic.

# Background

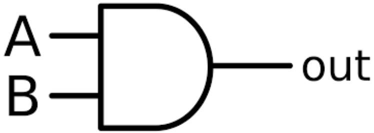
We know computer only understand binary bits ('0' and '1'). Let's think about a few basic mathematical operations.

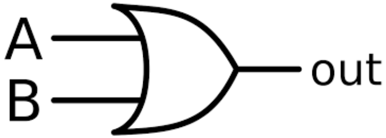
- Addition: We can do this operation using Half/Full adder
- Subtraction: Add with 2's Compliment
- Multiplication: Multiple time addition
- Division: Multiple-time subtraction

- Addition: 4 + 5 = 9 (0100 + 0101 = 1001)
- Subtraction: 10 – 9 = 1 (1010 + 0111 = 0001)
- Multiplication: 5x4 = 4+4+4+4+4 = 20
- Division: 10/2 = 2 can be subtracted from 10, 5 times

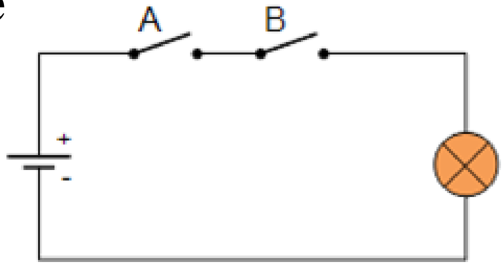
# Background

- In practice while implementing any logic function we need to know about Switching. Using different types of switches we can easily implement any logic function

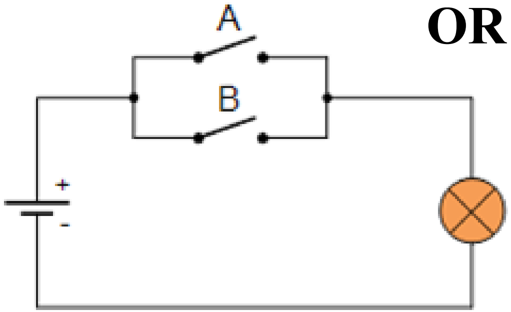



**AND Gate**




**OR Gate**




The faster you can operate these switches, the faster you can complete the functions!!!

# Background

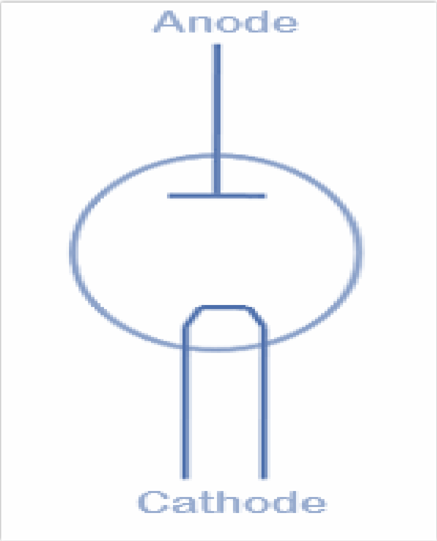
- The speed of computational operations depends on the ‘Speed of Switching’
- Normal Mechanical Switch: Not Reliable
- Causes: Bulky and heavy,
- Mechanical wear over time,
- Noisy, ultra-slow
- Requires lots of energy to operate
- Electronics: Help to increase the switching speed

SPST Switch

Electromechanical Relay

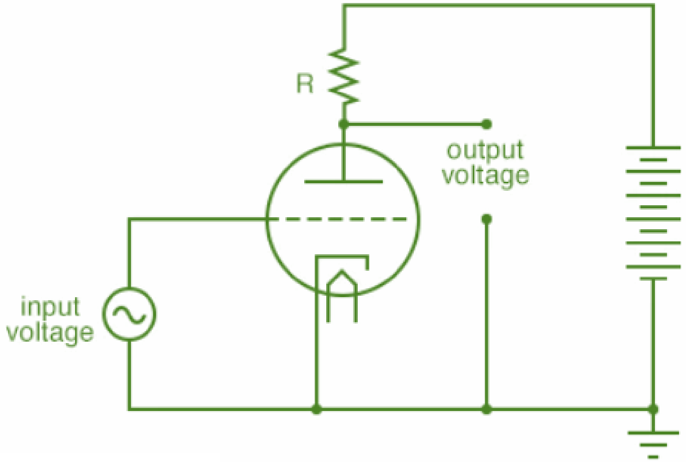
# First Electronic Switch

- The first electronic switch, the **vacuum tube (or thermionic valve)**, was invented by **John Ambrose Fleming** in **1904**.
- (-) ve terminal : Cathode
- (+) ve terminal : Anode
- Cathode was heated at high temperature, that emits electrons (<u>thermionic emission</u>) which travel to the anode, allowing current to flow in one direction



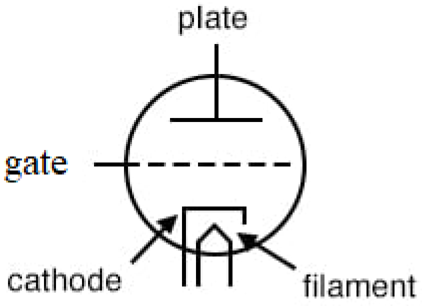
# First Electronic Switch

- In **1906**, **Lee De Forest** improved the previous configuration by adding a third electrode (**grid/gate**), which is placed between cathode and anode.
- By applying a small voltage to the grid, the flow of electrons can be controlled, allowing the tube to act as a **switch**.



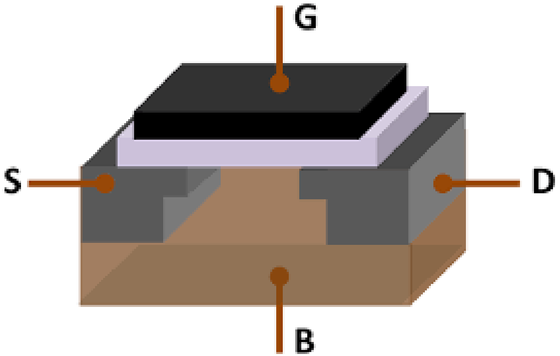

Disadvantage:

✓ High Voltage Needed
✓ Burn due to High Temperature (Non-reliable)
✓ Bulky



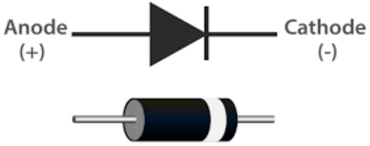
# Recent Electronic Switch

- No moving parts
- Scalable
- High speed

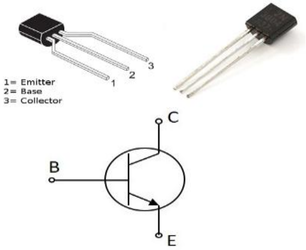



**5GHz computer → 5 billion operations per second !!!**

Any circuit consisting of semiconductors can be considered as electronic SW.




Diode




BJT


MOSFET

# CSE 251: Where does it fit in?

High level programming language ( C++, Java, Python etc.)

Assembly language (x86 assembly language, ARM, CUDA)

Machine Language

Architecture (RISC, CISC, IBM Power; Instruction sets varies depending on the architecture)

System level (Register, MUX, Encoder, Decoder)

Computer Architecture and Microprocessor courses

DLD Course

# CSE 251: Where does it fit in?

CSE 350 + CSE 460

Gate level (AND, OR, NOT) + Logic level + Logic Family

CSE 251

Transistors/ Switch (Digital Electronics)

# CSE 251: Where does it fit in?

**Digital Electronics**

- Boolean logic
- Addition, subtraction, multiplication, division

**Analog Electronics**

- Amplifiers, radio transmitter and receivers, modulator

**Power Electronics**

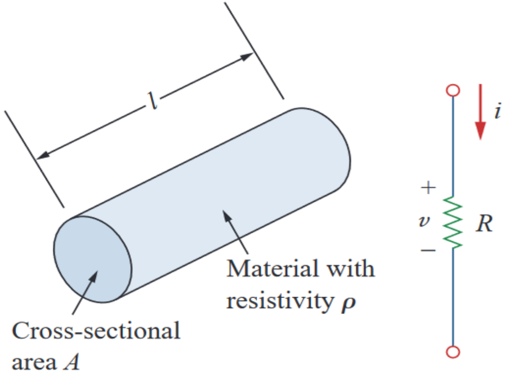
- Motor control
- AC to DC conversion or vice versa
- HVDC circuits
- Charge control circuits

# Revisit to CSE 250: Ohm's Law

The Fundamental *Ohm's Law*: The voltage across a resistor *v* is **directly proportional** to the current *i* flowing through the resistor.

$$v \propto i$$

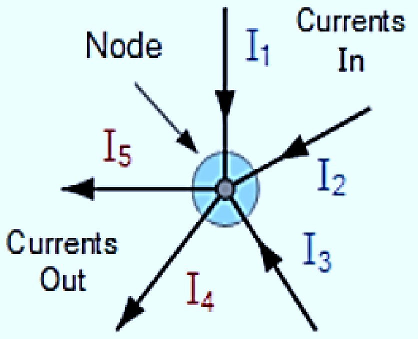
$$v = iR$$



# Revisit to CSE 250: KCL

**KCL: "Total current entering a junction or node is exactly equal to the charge leaving the node as it has no other place to go except to leave, as no charge is lost within the node"**

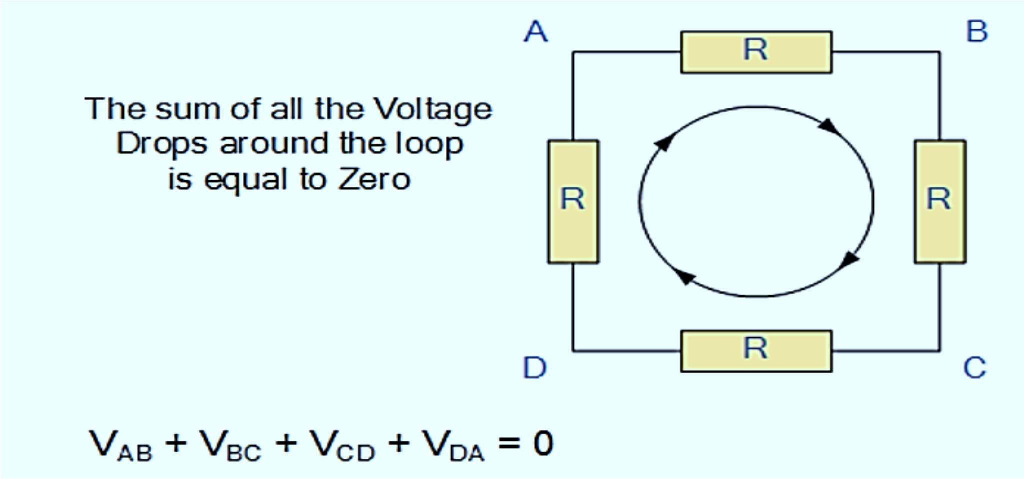
Currents Entering the Node
Equals
Currents Leaving the Node




$$I_1 + I_2 + I_3 + (-I_4 + -I_5) = 0$$

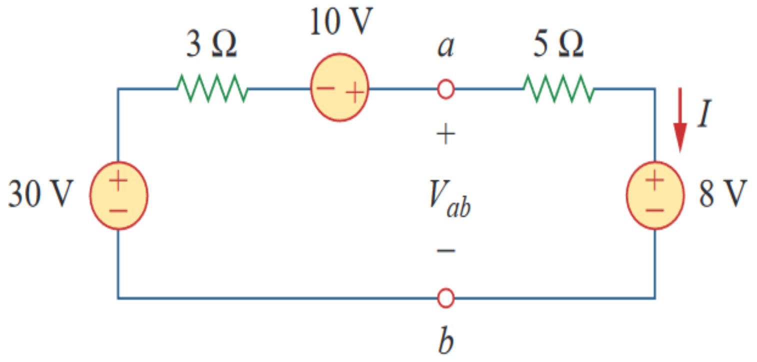
# Revisit to CSE 250: KVL

**KVL: "In any closed loop network, the total voltage around the loop is equal to the sum of all the voltage drops within the same loop"**



# Revisit to CSE 250: KVL - Example 1

Question: Find $I$ and $V_{ab}$ in the circuit




**Solution:**

KVL

$$-30 + 3I - 10 + 5I + 8 = 0$$

$$I = \frac{32}{8}\,\text{A} = 4\ \text{A}$$

KVL

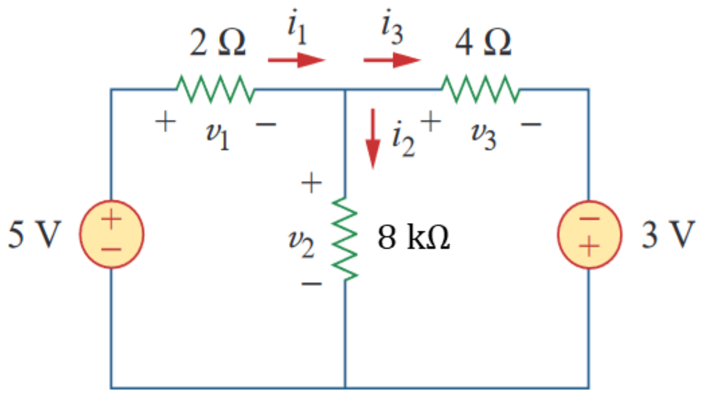
$$-V_{ab} + 5I + 8 = 0$$

$$V_{ab} = 28\ \text{V}$$

**Tip:** If you find resistance values in **kΩ** instead of **Ω**, don't convert the **kΩ** values to **Ω**. Just find currents in **mA** instead of **A**.

# Revisit to CSE 250: KVL - Example 2

Question: Find $v_1, v_2, v_3, i_1, i_2$ and $i_3$ in the circuit.




**Solution:**

KVL in first loop

$$-5 + 2i_1 + 8(i_1 - i_3) = 0$$

$$10i_1 - 8i_3 = 5$$

KVL in second loop

$$-8(i_1 - i_3) + 4i_3 - 3 = 0$$

$$-8i_1 + 12i_3 = 3$$

Solving:

$$i_1 = 1.5 \text{ mA}$$
$$i_3 = 1.25 \text{ mA}$$

$$v_1 = 3 \text{ V}$$
$$v_2 = 2 \text{ V}$$
$$v_3 = 5 \text{ V}$$

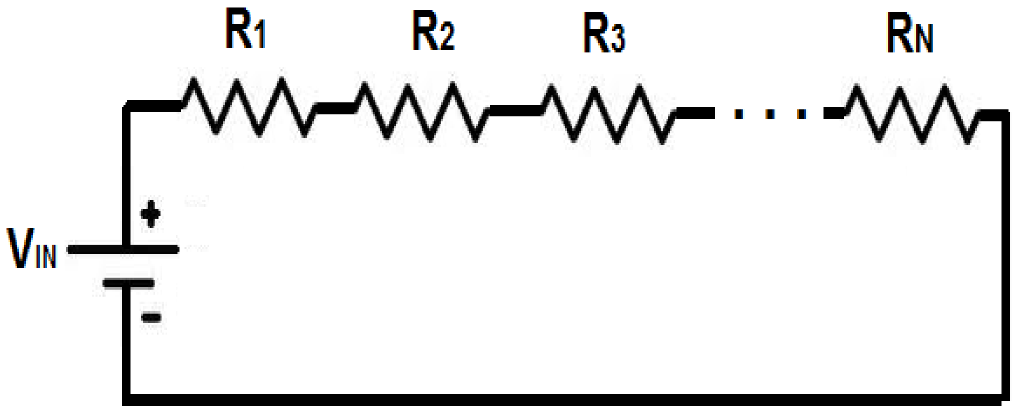
**Tip:** If you find resistance values in **kΩ** instead of **Ω**, don't convert the **kΩ** values to **Ω**. Just find currents in **mA** instead of **A**.

# Revisit to CSE 250: Series Resistors and Voltage Division

VDR: If there are ***N*** resistors in series, the voltage across the ***i*** –th resistor is given by,

$$R_{eq} = \sum_i R_i$$

$$v_i = \frac{\boldsymbol{R_i}}{\sum_i R_i} \boldsymbol{v_{in}}$$



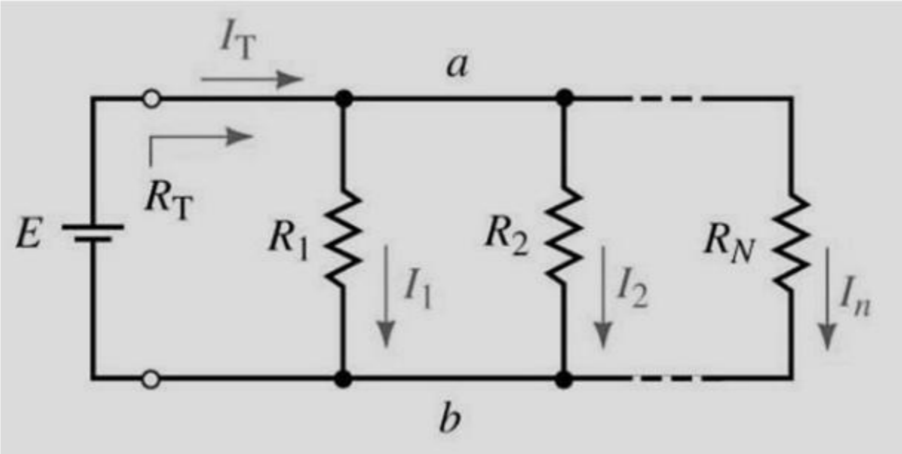
# Revisit to CSE 250: Series Resistors and Voltage Division

CDR: If there are $N$ resistors in parallel, the current through the $i$ – th resistor is given by, $i \in \{1,2,3,\cdots N\}$

As $I \propto \frac{1}{R}$

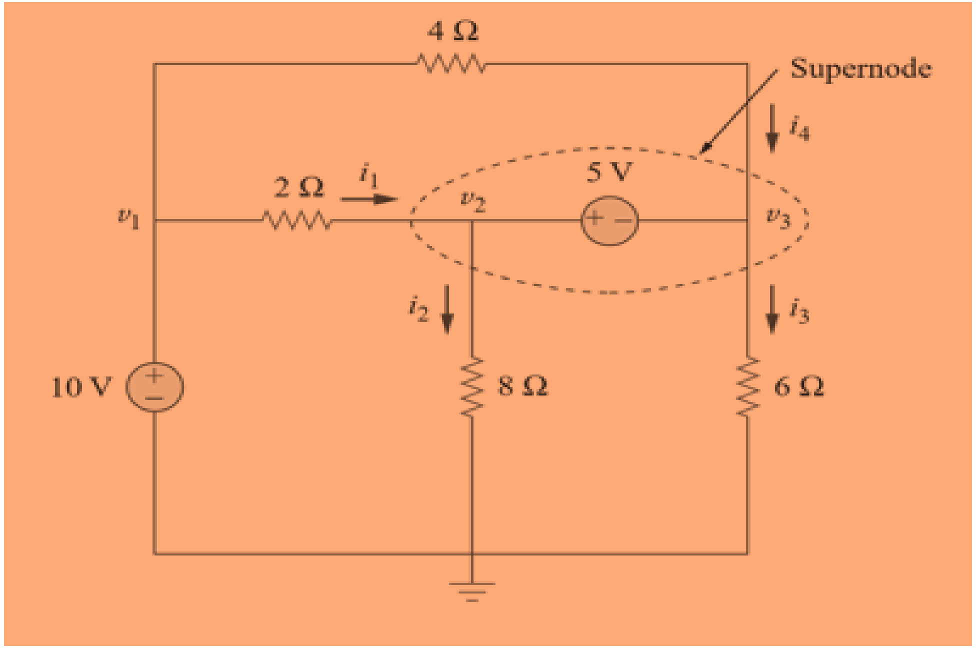
$$\frac{1}{R_{eq}} = \sum_i 1/R_i$$

$$I_i = \frac{1/R_i}{\sum_i 1/R_i} I_T$$



# Revisit to CSE 250: Nodal Analysis

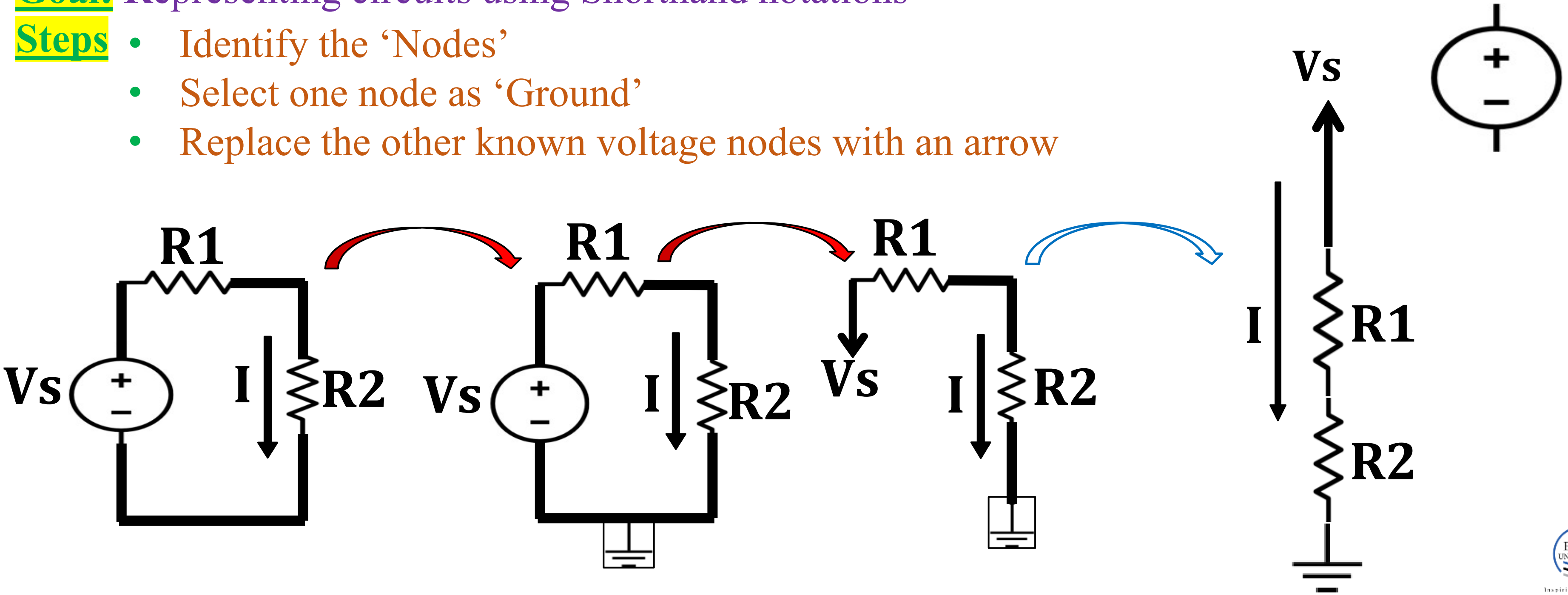
Write down the node equations for the given circuit.



# Alternative Circuit Representation: Line diagrams

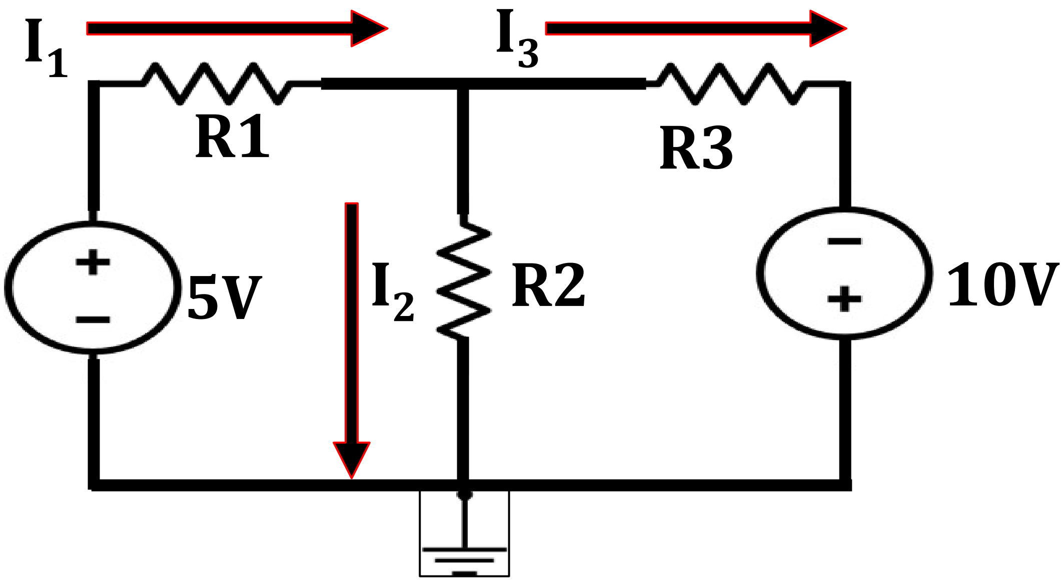
**Goal:** Representing circuits using Shorthand notations

**Steps**
- Identify the 'Nodes'
- Select one node as 'Ground'
- Replace the other known voltage nodes with an arrow

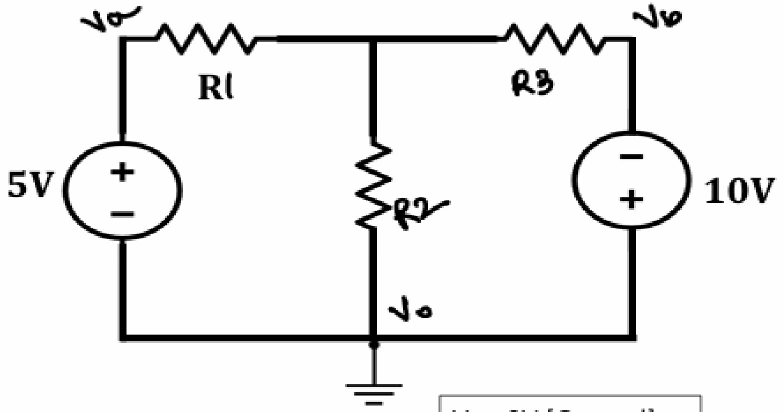


# Line diagrams: Example 1

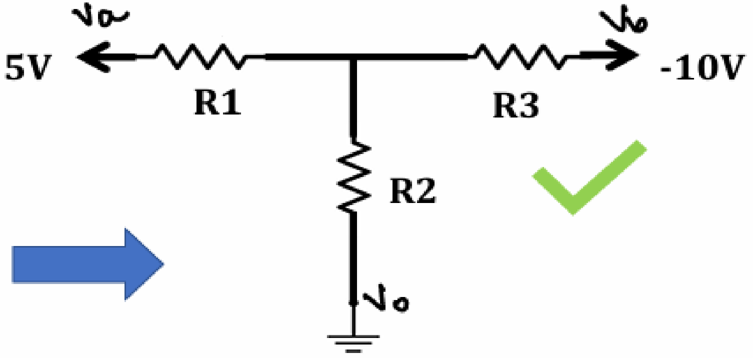
Question: Find the line diagram of the given circuit with voltage sources of opposite polarities.

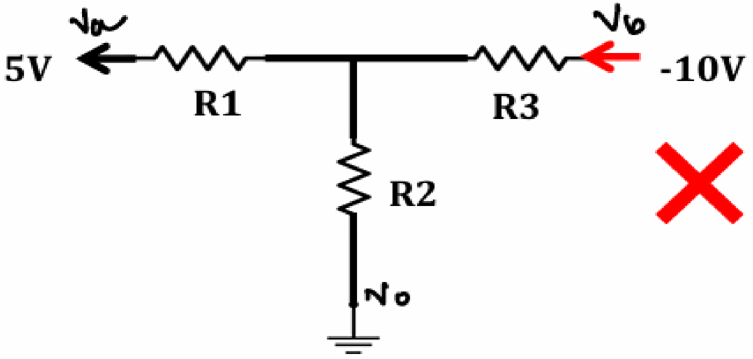


# Line diagrams: Example 1 (Solution)




Vo= 0V [Ground]
**For 5V :**
Va-Vo= (5-0) V
=> Va = 5V
**For 10V :**
Vo-Vb= (0-10) V
=> Vb = -10 V

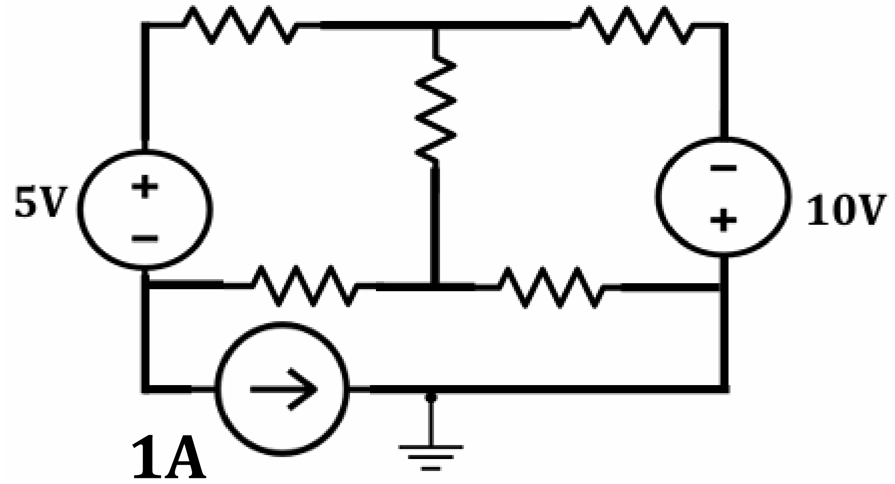


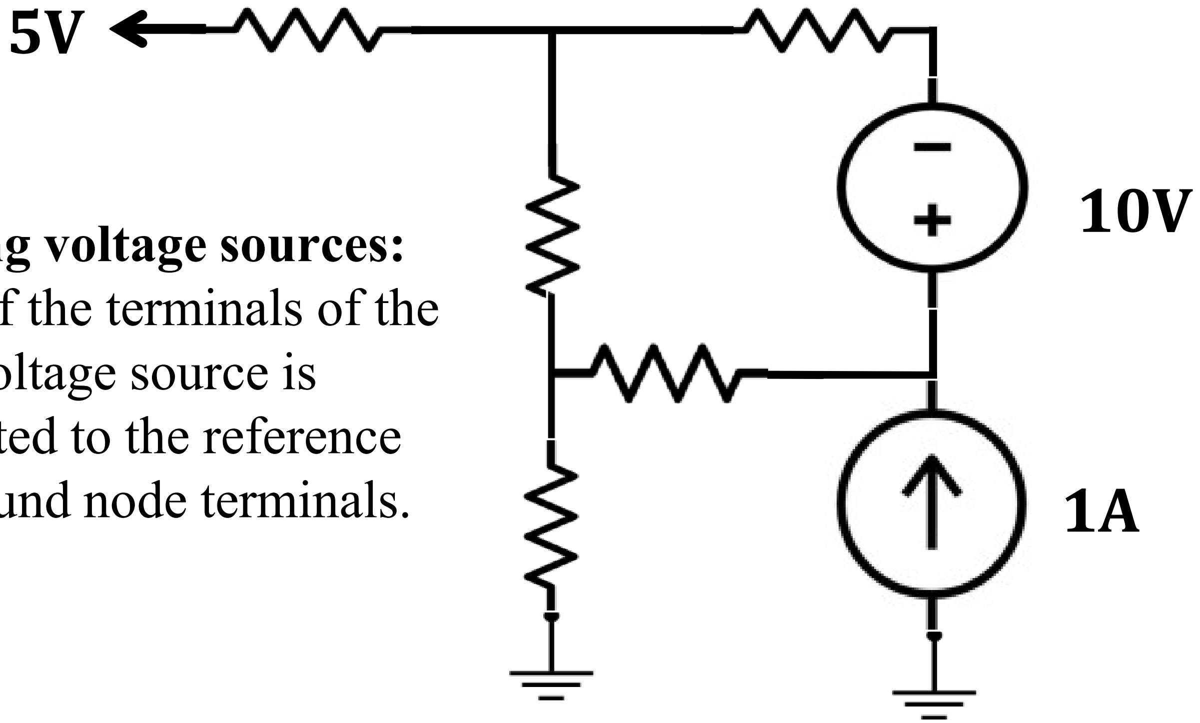
# Line diagrams: Example 2

What if the circuit contains a current source/ floating voltage source!!!!
Solution: Keep them as they are!

Question: Find the line diagram of the given circuit.

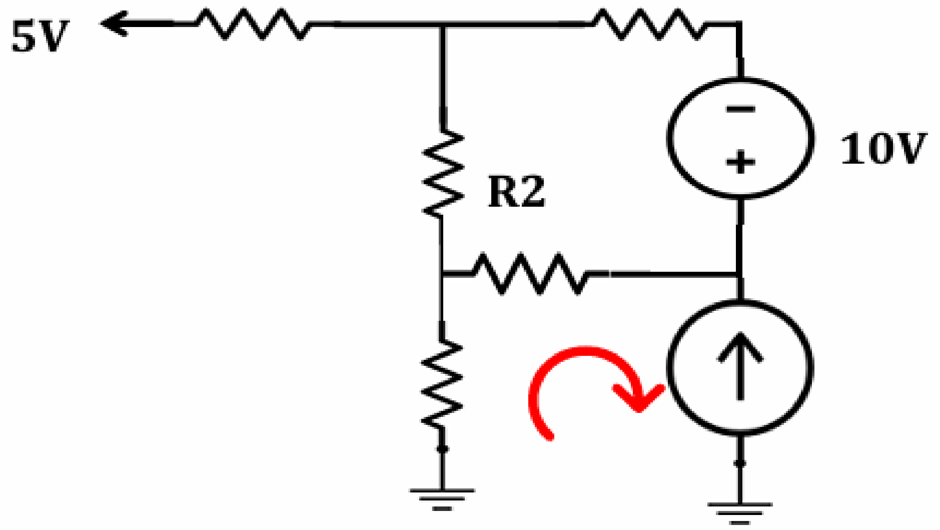


# Line diagrams: Example 2 (Solution)




**Floating voltage sources:** None of the terminals of the 10 V voltage source is connected to the reference i.e. ground node terminals.

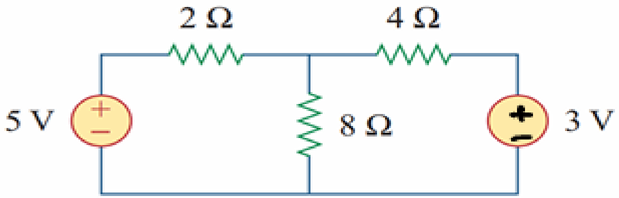
# Line diagrams

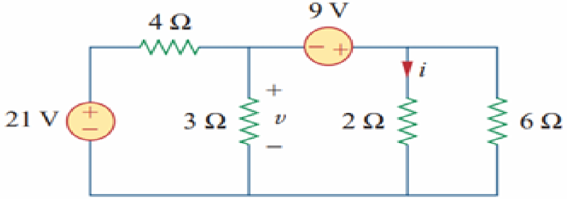


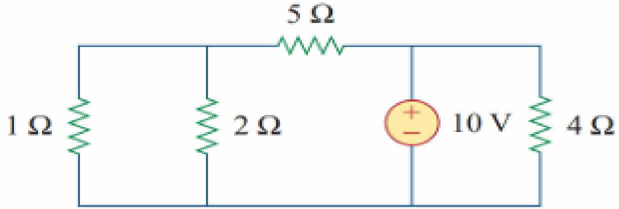

Can you write a KVL equation along this line?

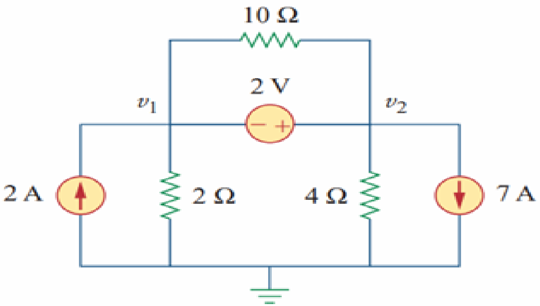
# Line diagrams: Practice Problems

Questions: i) Draw Alternative Circuit Diagrams, ii) Write down KCL equations, iii) Write down KVL equations, and iv) Nodal equation



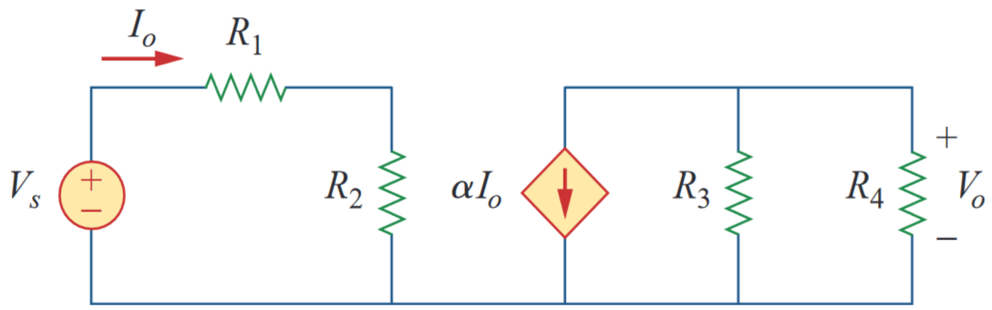



# Line diagrams: Example 3

Questions: (i) Find the alternative circuit representation of the following circuit.

(ii) For the circuit, find $\left|\frac{V_o}{V_s}\right|$ in terms of $\boldsymbol{\alpha, R_1, R_2, R_3}$ and $\boldsymbol{R_4}$.

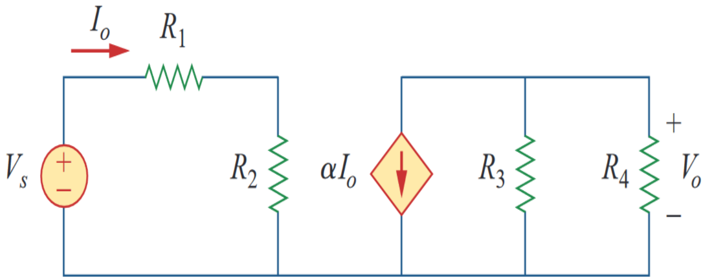
(iii) If $\boldsymbol{R_1} = \boldsymbol{R_2} = \boldsymbol{R_3} = \boldsymbol{R_4}$ what value of $\boldsymbol{\alpha}$ will produce $\left|\frac{V_o}{V_s}\right| = 10$?



# Line diagrams: Example 3 (Solution)

**Solution:**

Ohm's Law across $\boldsymbol{R_1} + \boldsymbol{R_2}$.




$$I_O = \frac{V_S}{R_1 + R_2}$$

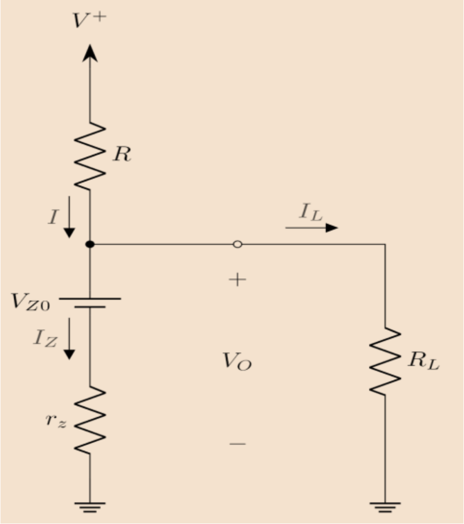
$$\boldsymbol{i} = -\boldsymbol{\alpha I_O}$$

Voltage across **Parallel Resistors** $\boldsymbol{R_3}, \boldsymbol{R_4}$

$$V_O = i(\boldsymbol{R_3}||\boldsymbol{R_4}) = -\frac{\alpha V_S}{R_1 + R_2} \cdot \frac{R_3 R_4}{R_3 + R_4} \qquad \left|\frac{V_O}{V_S}\right| = \frac{\alpha}{R_1 + R_2} \cdot \frac{R_3 R_4}{R_3 + R_4}$$

# Line diagrams: Practice Problem

Question: For the circuit given $\boldsymbol{R} = 100\ \Omega$, $\boldsymbol{R_L} = 10\ \text{k}\Omega$, $\boldsymbol{r_z} = 20\ \Omega$, $\boldsymbol{V_{ZO}} = 3$ V, and $\boldsymbol{I_Z} = 1$ mA. Find $\boldsymbol{V_O}$, $\boldsymbol{I_L}$, $\boldsymbol{I}$, $\boldsymbol{V^+}$.



# Line diagrams: Practice Problem

For more problems, please refer to the given link: [Test Your Understanding]

# Thank you for your attention